UJJAWAL PRATAP SINGH

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EDUCATION

University of Alberta, Faculty of Science, Edmonton, AB May 2027
B.Sc Computing Specialization Program
Awards: University of Alberta Regional Excellence Scholarship ($10,000)
Relevant Coursework: File and Database Management, Intro to Machine Learning, Reinforcement Learning, Computer Architecture, Web Applications and Architecture, Search and Planning in Artificial Intelligence, Data Structures & Algorithms

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, HTML/CSS, C, Bash, RISC-V
Web & Backend: React, Node.js, Django, Django REST Framework, FastAPI, TailwindCSS, REST APIs, Chrome Extension APIs
Machine Learning: RAG, NLP, PyTorch, Reinforcement Learning, TensorFlow, Neural Networks, MediaPipe
DevOps & Cloud: Docker, Heroku, Gunicorn, Git/GitHub, Linux/Unix, Firebase/GCP, Cloudinary
Data & Analytics: PostgreSQL, MongoDB (NoSQL), SQLite, Query Optimization, Data Engineering Pipelines

PROJECTS

Leo the Lion | AI-Powered Pediatric Health Platform (Hackathon Top 5)

- **Full-Stack Architecture:** Architected a real-time health companion using **React 18 (Frontend)** and **FastAPI (Backend)**, featuring a dual-interface system for children and parents.
- **AI & RAG Integration:** Engineered a **Retrieval-Augmented Generation (RAG)** engine using **Python** and Sentence Transformers to ground responses in medical protocols, utilizing **Google Gemma 3** for persona-driven dialogue.
- **Computer Vision & Voice:** Implemented real-time object detection using **TensorFlow.js (Coco-SSD)** for interactive play and a custom **Web Speech API integration** for an immersive "Leo" voice persona.
- **Real-Time Optimization:** Engineered a low-latency bi-directional streaming architecture using **Server-Sent Events (SSE)**, reducing perceived latency by 40% for instant voice and text interaction.

WebAble | AI-Powered Accessibility Extension (JavaScript, MediaPipe, Neural Networks)

- **Local Neural Network Inference:** Engineered a zero-dependency Multi-Layer Perceptron (MLP) inference engine in vanilla JavaScript, performing real-time forward propagation and softmax classification on 63-dimensional data to predict American Sign Language (ASL) gestures locally.
- **Real-Time Computer Vision:** Integrated Google MediaPipe Hands to extract 3D landmarks via optimized web-cam and screen-sharing capture loops, achieving high-accuracy spatial normalization at 60 FPS.
- **Extension Architecture & Security:** Architected a secure, cross-context browser extension using isolated iframes (asl-frame), background script messaging, and dynamic DOM injection to render accessibility layers (like Dyslexia reading modes) while bypassing strict Content Security Policies (CSP).

Social Distribution | Federated Social Network (Django, DRF, PostgreSQL, Heroku)

- **Federated Inbox Architecture:** Engineered a node-to-node federation layer delivering entries, likes, comments, and follow requests across independently deployed nodes via an inbox API, authenticated with HTTP Basic Auth and guarded by a custom DRF authentication backend**.**
- **Visibility & Access Control:** Designed a three-tier visibility model (public, unlisted, friends-only) with bidirectional follow detection, enforcing friends-only access behind mutual-follow checks at both the view and serializer layers.
- **Production Deployment & Integrations:** Shipped to Heroku with PostgreSQL, Gunicorn, and WhiteNoise; integrated Cloudinary for image hosting and the GitHub Events API to auto-ingest user activity as markdown entries sanitized via markdown-it-py and bleach.

LEADERSHIP

Indian Students Association (INDSA), VP Media Jan 2025 - Sep 2025

- **GenAI Integration:** Integrated **Generative AI workflows** (LLMs, Image Gen) into the content strategy to accelerate drafting and design processes, reducing production time while maintaining brand consistency.
- **Data-Driven Optimization:** Analyzed audience engagement metrics using **AI-powered analytics tools** to identify trends and optimize scheduling, directly resulting in a sold-out major cultural event.